Personal Trainer KSA

Phase 3

NAME	ID
[redacted]	[redacted]
Lama[redacted]	[redacted]
[redacted]	[redacted]
Ebtisam Bassam Alotaibi	[redacted]
[redacted]	4[redacted]

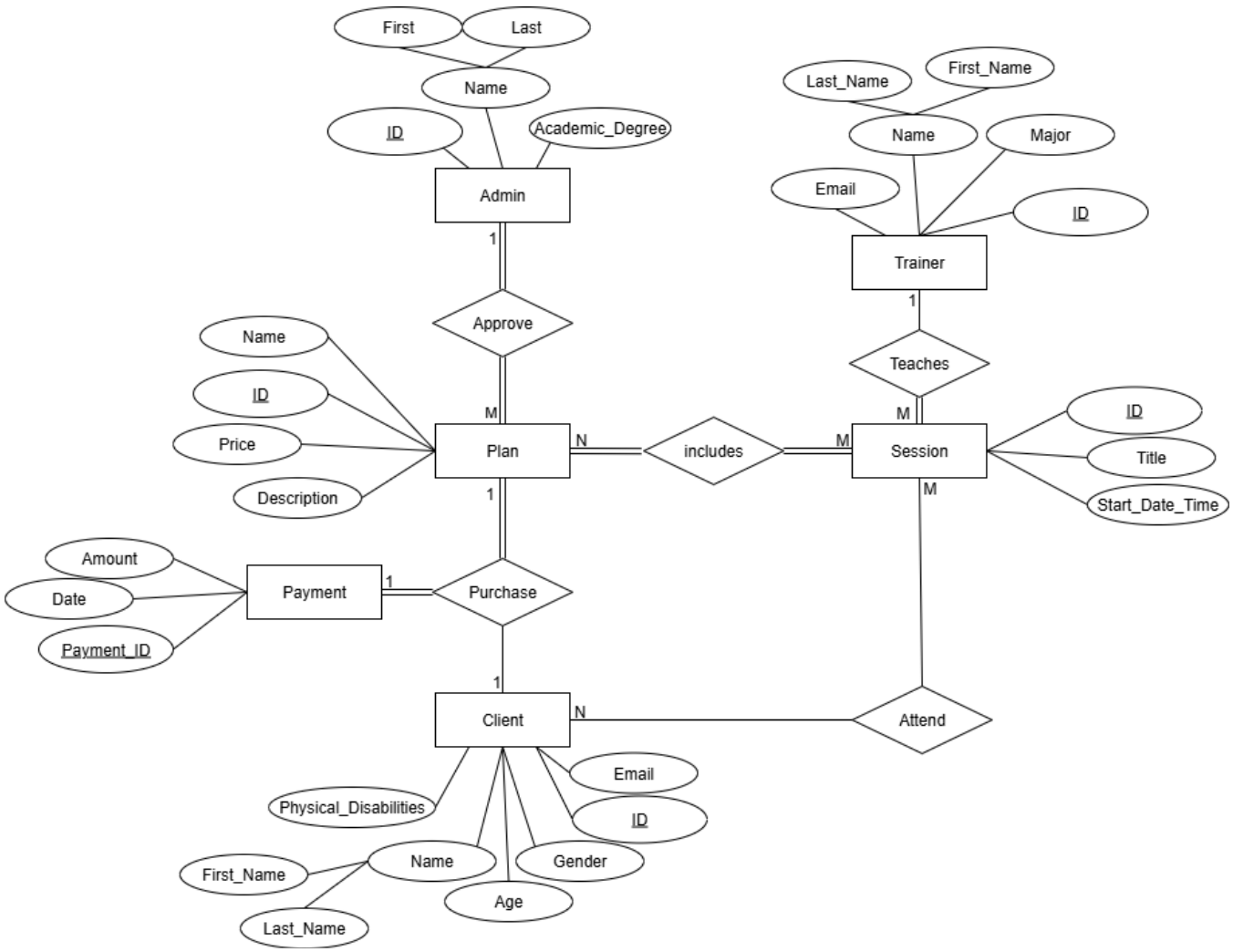
Section#: 3C4
Group#: 2

User Identification and Tasks

The project is a system for managing training plans and sessions, where clients can register, choose plans, and make payments, trainers provide sessions, and admins approve plans before publishing.

Users	*Activities or Tasks*
Client	▪ *Register an account in the system.* ▪ *Browse and select available training plans.* ▪ *Book training sessions.* ▪ *Make online payments.* ▪ *View their session schedule and trainer information.*
Trainer	▪ *Create and manage training sessions.* ▪ *Update session details (date, time, capacity).* ▪ *Track client attendance and progress.* ▪ *Communicate with clients through the system.*
Admin	▪ *Approve and publish training plans created by trainers.* ▪ *Manage system users (clients and trainers).* ▪ *Handle payment verification and report generation.* ▪ *Monitor system activities for accuracy and consistency.*

Entity Relationship Diagram (ERD):



Section 3:Relational Model

In this section, the ERD is mapped into relational tables.
Each table includes its attributes,primary kay,and foreign keys:

Admin(Admin_ID,First_Name,Last_Name,Academic_Degree)

Plan(Plan_ID,Name,Price,Description,Admin_ID)

FK: Admin_ID references Admin (Admin_ID)

Payment(Payment_ID,Date ,Amount)

Client(Client_ID,Email,Gender,Age,Physical_Disabilities,First_Name,Last_Name)

Client-plan(Client_ID, Plan_ID, Payment_ID)

FK1: Client_ID references Client(Client_ID)

FK2: Plan_ID references Plan(Plan_ID)

FK3: Payment_ID references Payment(Payment_ID)

Trainer(Trainer_ID,Email,Major,First_Name,Last_Name)

Session(Session_ID,Title,Start_Date_Time,Trainer_ID)

FK1: Trainer_ID references Trainer(Trainer_ID)

Plan-Session(Plan_ID,Session_ID)

FK1: Plan_ID references Plan(Plan_ID)

FK2: Session_ID references Session(Session_ID)

Client-Session(Client_ID, Session_ID)

FK1: Client_ID references Client(Client_ID)

FK2: Session_ID references Session(Session_ID)

Implementation phase

Submit the COMMANDS and the resulted OUTPUT (snapshot) from the execution of the DB table's creation, data insertion and the data queries.

1. Create Database Tables

```
CREATE TABLE Admin (
    Admin_ID NUMBER PRIMARY KEY,
    First_Name VARCHAR2(50) NOT NULL,
    Last_Name VARCHAR2(50) NOT NULL,
    Academic_Degree VARCHAR2(100)
);

CREATE TABLE Plan (
    Plan_ID NUMBER PRIMARY KEY,
    Name VARCHAR2(50) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Description VARCHAR2(255),
    Admin_ID NUMBER,
    FOREIGN KEY (Admin_ID) REFERENCES Admin(Admin_ID)
);

CREATE TABLE Payment (
    Payment_ID NUMBER PRIMARY KEY,
    Payment_Date DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL
);

CREATE TABLE Client (
    Client_ID NUMBER PRIMARY KEY,
    Email VARCHAR2(80) NOT NULL UNIQUE,
    Gender CHAR(1) CHECK (Gender IN ('M', 'F')),
    Age NUMBER CHECK (Age >= 0),
    Physical_Disabilities VARCHAR(200),
    First_Name VARCHAR2(50) NOT NULL,
    Last_Name VARCHAR2(50) NOT NULL
);
```

```
CREATE TABLE Plan_Session (
    Plan_ID NUMBER,
    Session_ID NUMBER,
    PRIMARY KEY (Plan_ID, Session_ID),
    FOREIGN KEY (Plan_ID) REFERENCES Plan(Plan_ID),
    FOREIGN KEY (Session_ID) REFERENCES Session_Table(Session_ID)
);

CREATE TABLE Client_Session (
    Client_ID NUMBER,
    Session_ID NUMBER,
    PRIMARY KEY (Client_ID, Session_ID),
    FOREIGN KEY (Client_ID) REFERENCES Client(Client_ID),
    FOREIGN KEY (Session_ID) REFERENCES Session_Table(Session_ID)
);
```

```sql
CREATE TABLE Client_Plan (
    Client_ID NUMBER,
    Plan_ID NUMBER,
    Payment_ID NUMBER,
    PRIMARY KEY (Client_ID, Plan_ID, Payment_ID),
    FOREIGN KEY (Client_ID) REFERENCES Client(Client_ID),
    FOREIGN KEY (Plan_ID) REFERENCES Plan(Plan_ID),
    FOREIGN KEY (Payment_ID) REFERENCES Payment(Payment_ID)
);

CREATE TABLE Trainer (
    Trainer_ID NUMBER PRIMARY KEY,
    Email VARCHAR2(80) NOT NULL UNIQUE,
    Major VARCHAR2(100) NOT NULL,
    First_Name VARCHAR2(50) NOT NULL,
    Last_Name VARCHAR2(50) NOT NULL
);

CREATE TABLE Session_Table (
    Session_ID NUMBER PRIMARY KEY,
    Title VARCHAR2(100) NOT NULL,
    Start_Date_Time TIMESTAMP NOT NULL,
    Trainer_ID NUMBER NOT NULL,

    FOREIGN KEY (Trainer_ID) REFERENCES Trainer(Trainer_ID)
);
```

```
Script Output
Task completed in 0.737 seconds

Table ADMIN created.

Table PLAN created.

Table PAYMENT created.

Table CLIENT created.

Table CLIENT_PLAN created.

Table TRAINER created.

Table SESSION_TABLE created.

Table PLAN_SESSION created.

Table CLIENT_SESSION created.
```

```
Welcome Page | Drop.sql | Create_Tables.sql | Insert.sql | Query.sql
SQL Worksheet | History
Worksheet | Query Builder
DROP TABLE Admin CASCADE CONSTRAINTS;

DROP TABLE Plan CASCADE CONSTRAINTS;

DROP TABLE Payment CASCADE CONSTRAINTS;

DROP TABLE Client CASCADE CONSTRAINTS;

DROP TABLE Client_Plan CASCADE CONSTRAINTS;

DROP TABLE Trainer CASCADE CONSTRAINTS;

DROP TABLE Session_Table CASCADE CONSTRAINTS;

DROP TABLE Plan_Session CASCADE CONSTRAINTS;

DROP TABLE Client_Session CASCADE CONSTRAINTS;
```

```
Script Output
Task completed in 0.832 seconds

Table ADMIN dropped.

Table PLAN dropped.

Table PAYMENT dropped.

Table CLIENT dropped.

Table CLIENT_PLAN dropped.

Table TRAINER dropped.

Table SESSION_TABLE dropped.

Table PLAN_SESSION dropped.

Table CLIENT_SESSION dropped.
```

2. Insert Data

```
-- Admins (5)
INSERT INTO Admin VALUES (1, 'Sara', 'Al-Qahtani', 'MSc Information Systems');
INSERT INTO Admin VALUES (2, 'Nora', 'Al-Harbi', 'PhD Computer Science');
INSERT INTO Admin VALUES (3, 'Laila', 'Al-Mutairi', 'MSc Management');
INSERT INTO Admin VALUES (4, 'Maha', 'Al-Ahmadi', 'MBA');
INSERT INTO Admin VALUES (5, 'Huda', 'Al-Otaibi', 'BSc Information Systems');
SELECT * FROM Admin;
COMMIT;

-- Plans (5) — اربطنا كل خطة بـ Admin
INSERT INTO Plan VALUES (301, 'Basic Fit', 150.00, 'Access to gym + 2 group sessions per week', 1);
INSERT INTO Plan VALUES (302, 'Premium Fit', 300.00, 'Full access + unlimited classes', 2);
INSERT INTO Plan VALUES (303, 'Yoga Special', 180.00, 'Weekly yoga classes + mat', 1);
INSERT INTO Plan VALUES (304, 'Cardio Pack', 120.00, 'Cardio classes 3x week', 3);
INSERT INTO Plan VALUES (305, 'Home Workout', 90.00, 'Online recorded sessions', 4);
SELECT * FROM Plan;
COMMIT;

-- Payments (5)
INSERT INTO Payment VALUES (501, DATE '2025-10-01', 150.00);
INSERT INTO Payment VALUES (502, DATE '2025-10-05', 300.00);
INSERT INTO Payment VALUES (503, DATE '2025-10-10', 180.00);
INSERT INTO Payment VALUES (504, DATE '2025-10-12', 120.00);
INSERT INTO Payment VALUES (505, DATE '2025-10-15', 90.00);
SELECT * FROM Payment;
COMMIT;

-- Clients (5)
INSERT INTO Client VALUES (1, 'lina@example.com', 'F', 23, 'None', 'Lina', 'AlSayed');
INSERT INTO Client VALUES (2, 'mona123@example.com', 'F', 27, NULL, 'Mona', 'AlShammari');
INSERT INTO Client VALUES (3, 'ahmad99@example.com', 'M', 30, 'Visual impairment', 'Ahmad', 'AlHarbi');
INSERT INTO Client VALUES (4, 'sara_fit@example.com', 'F', 21, 'None', 'Sara', 'AlQahtani');
INSERT INTO Client VALUES (5, 'fahad2025@example.com', 'M', 35, NULL, 'Fahad', 'AlOtaibi');
SELECT * FROM Client;
COMMIT;
```

```sql
--Client_Plan (5)
INSERT INTO Client_Plan VALUES (1, 301, 501);
INSERT INTO Client_Plan VALUES (2, 302, 502);
INSERT INTO Client_Plan VALUES (3, 303, 503);
INSERT INTO Client_Plan VALUES (3, 304, 504);
INSERT INTO Client_Plan VALUES (5, 305, 505);
SELECT * FROM Client_Plan;
COMMIT;

-- Trainers (5)
INSERT INTO Trainer VALUES (201, 'hala.trainer@example.com', 'Fitness and Conditioning', 'Hala', 'Al-Mutairi');
INSERT INTO Trainer VALUES (202, 'reem.yoga@example.com', 'Yoga and Meditation', 'Reem', 'Al-Saadi');
INSERT INTO Trainer VALUES (203, 'majid.cardio@example.com', 'Cardio and HIIT', 'Majid', 'Al-Harbi');
INSERT INTO Trainer VALUES (204, 'noor.strength@example.com', 'Strength Training', 'Noor', 'Al-Jahani');
INSERT INTO Trainer VALUES (205, 'samah.pilates@example.com', 'Pilates and Flexibility', 'Samah', 'Al-Otaibi');
SELECT * FROM Trainer;
COMMIT;
```

```sql
-- Session_Table (5)
-- 1) Hala — Fitness and Conditioning
INSERT INTO Session_Table
VALUES (401, 'Full Body Conditioning',
        TO_TIMESTAMP('2025-11-24 18:00:00', 'YYYY-MM-DD HH24:MI:SS'),
        201);
-- 2) Reem — Yoga and Meditation
INSERT INTO Session_Table
VALUES (402, 'Morning Yoga Flow',
        TO_TIMESTAMP('2025-11-20 08:00:00', 'YYYY-MM-DD HH24:MI:SS'),
        202);
-- 3) Majid — Cardio and HIIT
INSERT INTO Session_Table
VALUES (403, 'HIIT Cardio Blast',
        TO_TIMESTAMP('2025-11-21 17:00:00', 'YYYY-MM-DD HH24:MI:SS'),
        203);
-- 4) Noor — Strength Training
INSERT INTO Session_Table
VALUES (404, 'Strength Training Basics',
        TO_TIMESTAMP('2025-11-22 10:30:00', 'YYYY-MM-DD HH24:MI:SS'),
        204);
-- 5) Samah — Pilates and Flexibility
INSERT INTO Session_Table
VALUES (405, 'Pilates Flex Session',
        TO_TIMESTAMP('2025-11-23 09:00:00', 'YYYY-MM-DD HH24:MI:SS'),
        205);
SELECT * FROM Session_Table;
COMMIT;

-- Plan_Session (5)
INSERT INTO Plan_Session VALUES (301, 401);
INSERT INTO Plan_Session VALUES (302, 402);
INSERT INTO Plan_Session VALUES (303, 405);
INSERT INTO Plan_Session VALUES (304, 403);
INSERT INTO Plan_Session VALUES (305, 404);
SELECT * FROM Plan_Session;
COMMIT;
```

```sql
-- Client_Session (5)
INSERT INTO Client_Session VALUES (1, 401);
INSERT INTO Client_Session VALUES (2, 402);
INSERT INTO Client_Session VALUES (3, 403);
INSERT INTO Client_Session VALUES (4, 404);
INSERT INTO Client_Session VALUES (5, 405);
SELECT * FROM Client_Session;
COMMIT;
```

```
Script Output
Task completed in 1.241 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

 ADMIN_ID FIRST_NAME                                        LAST_NAME                                        ACADEMIC_DEGREE
---------- ---------------------------------------------------- ---------------------------------------------------- ---------------------------------------------------
         1 Sara                                               Al-Qahtani                                       MSc Information Systems
         2 Nora                                               Al-Harbi                                         PhD Computer Science
         3 Laila                                              Al-Mutairi                                       MSc Management
         4 Maha                                               Al-Ahmadi                                        MBA
         5 Huda                                               Al-Otaibi                                        BSc Information Systems

Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.
```

```
Script Output
Task completed in 1.241 seconds

1 row inserted.

1 row inserted.

   PLAN_ID NAME                                               PRICE DESCRIPTION
---------- ---------------------------------------------------- ------ ----------------------------------------------------------------------------------------------------
       301 Basic Fit                                            150 Access to gym + 2 group sessions per week
       302 Premium Fit                                          300 Full access + unlimited classes
       303 Yoga Special                                         180 Weekly yoga classes + mat
       304 Cardio Pack                                          120 Cardio classes 3x week
       305 Home Workout                                          90 Online recorded sessions

Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

PAYMENT_ID PAYMENT_D     AMOUNT
---------- --------- ---------
```

```
Script Output
Task completed in 1.241 seconds

PAYMENT_ID PAYMENT_D     AMOUNT
---------- --------- ---------
       501 01-OCT-25       150
       502 05-OCT-25       300
       503 10-OCT-25       180
       504 12-OCT-25       120
       505 15-OCT-25        90

Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

 CLIENT_ID EMAIL                                                G       AGE PHYSICAL_DISABILITIES
---------- ---------------------------------------------------- - --------- ----------------------------------------
         1 lina@example.com                                     F        23 None
         2 mona123@example.com                                  F        27
         3 ahmad99@example.com                                  M        30 Visual impairment
         4 sara_fit@example.com                                 F        21 None
         5 fahad2025@example.com                                M        35
```

Script Output

Task completed in 1.241 seconds

```
Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

 CLIENT_ID    PLAN_ID PAYMENT_ID
---------- --------- ----------
         1       301        501
         2       302        502
         3       303        503
         3       304        504
         5       305        505

Commit complete.

1 row inserted.

1 row inserted.
```

Script Output

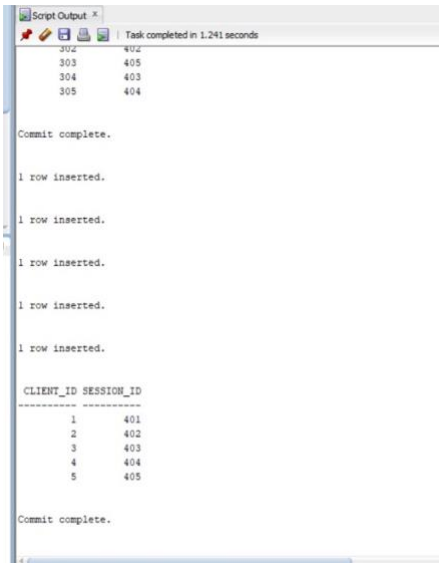
Task completed in 1.241 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

TRAINER_ID EMAIL                                                                 MAJOR
---------- ------------------------------------------------------------------------ ------------------------------------------------------------
       201 hala.trainer@example.com                                                 Fitness and Conditioning
       202 reem.yoga@example.com                                                    Yoga and Meditation
       203 majid.cardio@example.com                                                 Cardio and HIIT
       204 noor.strength@example.com                                                Strength Training
       205 samah.pilates@example.com                                                Pilates and Flexibility

Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Script Output

Task completed in 1.241 seconds

```
SESSION_ID TITLE                                                                 START_DATE_TIME                TRAINER_
---------- ------------------------------------------------------------------------ ------------------------------ --------
       401 Full Body Conditioning                                                   24-NOV-25 06.00.00.000000000 PM        2
       402 Morning Yoga Flow                                                        20-NOV-25 08.00.00.000000000 AM        2
       403 HIIT Cardio Blast                                                        21-NOV-25 05.00.00.000000000 PM        2
       404 Strength Training Basics                                                 22-NOV-25 10.30.00.000000000 AM        2
       405 Pilates Flex Session                                                     23-NOV-25 09.00.00.000000000 AM        2

Commit complete.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

   PLAN_ID SESSION_ID
---------- ----------
       301        401
       302        402
       303        405
       304        403
       305        404
```



3. Write and Execute SQL Queries

Scenario:

1.JOIN:
Scenario: The **Training Manager** requested a report showing the names of all **Trainers** and the **Titles of the Sessions** each one taught, along with the dates, to evaluate workload and session distribution.

2. GROUP BY & SUM
Scenario: The **Accounting Department** requested the **total amount paid by each client** since registration, to identify high-value clients eligible for special loyalty discounts.

3. GROUP BY & HAVING
Scenario: The **Administrative Manager** requested a list of **Trainer IDs** who have **more than one session** registered in the system, to ensure fair distribution of training classes.

4. Nested Query
Scenario: The **Quality Assurance team** requested a list of **Plans and their Prices** that were approved and published by **Admins** holding a **PhD** degree, to highlight plans that underwent the highest level of academic review.

5. UNION
Scenario: The **Marketing Team** requested a **unified list** of all **email addresses** registered in the system (for both clients and trainers), to send a consolidated newsletter without duplication.

6. Aggregate Function (COUNT)

Scenario: The **System Manager** requested a quick and immediate statistic showing the **total number of Clients** registered in the system, for the purpose of preparing daily client growth reports.

Command:

```sql
--  Shows each trainer and the session they teach
SELECT T.First_Name , T.Last_Name, S.Title , S.start_date_time
FROM Trainer T , Session_Table S
WHERE T.Trainer_ID = S.Trainer_ID;


--  Calculates the total amount paid by each client
SELECT
    cp.Client_ID, SUM(p.amount) AS TotalPayments
FROM Client_Plan cp,  Payment p
WHERE cp.Payment_ID = p.Payment_ID
GROUP BY  cp.Client_ID;

--  Finds trainers who teach more than one session
SELECT
    Trainer_ID,
    COUNT(Session_ID) AS TotalSessions
FROM Session_Table
GROUP BY Trainer_ID
HAVING COUNT(Session_ID) > 1;


--SUBQUERY:  Retrieves plans managed by an admin with a PhD in Computer Science.
SELECT *
FROM Plan
WHERE Admin_ID IN (SELECT Admin_ID FROM Admin WHERE Academic_Degree = 'PhD Computer Science');


--UNION: Combines clients' and trainers' emails without duplicates.
SELECT Email FROM Client
UNION
SELECT Email FROM Trainer;
```

```sql
--  Counts all clients in the system
SELECT COUNT(*) AS NUM_OF_CLIENTS
FROM Client;


--UPDATE: Updates the disability info of client 2
UPDATE Client
SET Physical_Disabilities = 'Visual impairment'
WHERE Client_ID = 2;

SELECT * FROM Client;


--CANCEL: Removes client 1 from session 401
DELETE FROM Client_Session
WHERE Client_ID = 1 AND Session_ID = 401;

SELECT * FROM Client_Session;
```

```
-- The admin requested a report to show each client along with the plan they subscribed to in order to analyze which plans are most popular.
SELECT
    c.First_Name AS Client_FirstName,
    c.Last_Name AS Client_LastName,
    p.Name AS Plan_Name,
    p.Price
FROM Client c
INNER JOIN Client_Plan cp ON c.Client_ID = cp.Client_ID
INNER JOIN Plan p ON cp.Plan_ID = p.Plan_ID;

-- The customer support team requested a list of clients whose email contains the word "example" to verify the registration of a new batch of users.
SELECT
    Client_ID,
    First_Name,
    Last_Name,
    Email
FROM Client
WHERE Email LIKE '%example%';
```

Output:

```
Script Output
Task completed in 0.546 seconds

FIRST_NAME                                         LAST_NAME                                          TITLE
-------------------------------------------------- -------------------------------------------------- --------------------------------------------------
Hala                                               Al-Mutairi                                         Full Body Conditioning
Reem                                               Al-Saadi                                           Morning Yoga Flow
Majid                                              Al-Harbi                                           HIIT Cardio Blast
Noor                                               Al-Jahani                                          Strength Training Basics
Samah                                              Al-Otaibi                                          Pilates Flex Session


 CLIENT_ID TOTALPAYMENTS
---------- -------------
         1           150
         2           300
         3           300
         5            90

no rows selected

   PLAN_ID NAME                                                    PRICE DESCRIPTION
---------- -------------------------------------------------- --------- --------------------------------------------------------------------------------
       302 Premium Fit                                              300 Full access + unlimited classes
```

```
Script Output
Task completed in 0.546 seconds

EMAIL
--------------------------------------------------------------------------------
lina@example.com
mona123@example.com
ahmad99@example.com
sara_fit@example.com
fahad2025@example.com
hala.trainer@example.com
reem.yoga@example.com
majid.cardio@example.com
noor.strength@example.com
samah.pilates@example.com

10 rows selected.


NUM_OF_CLIENTS
--------------
             5


1 row updated.


 CLIENT_ID EMAIL                                                                G        AGE PHYSICAL_DISABILITIES
---------- -------------------------------------------------------------------- - ---------- ------------------------------------
         1 lina@example.com                                                     F         23 None
         2 mona123@example.com                                                  F         27 Visual impairment
         3 ahmad99@example.com                                                  M         30 Visual impairment
         4 sara_fit@example.com                                                 F         21 None
         5 fahad2025@example.com                                                M         35
```

```
1 row deleted.


 CLIENT_ID SESSION_ID
---------- ----------
         2        402
         3        403
         4        404
         5        405
```

CLIENT_FIRSTNAME	CLIENT_LASTNAME
Lina	AlSayed
Mona	AlShammari
Ahmad	AlHarbi
Ahmad	AlHarbi
Fahad	AlOtaibi

PLAN_NAME	PRICE
Basic Fit	150
Premium Fit	300
Yoga Special	150
Cardio Pack	120
Home Workout	90

CLIENT_ID	FIRST_NAME	LAST_NAME	EMAIL
1	Lina	AlSayed	lina@example.com
2	Mona	AlShammari	mona123@example.com
3	Ahmad	AlHarbi	ahmad99@example.com
4	Sara	AlQahtani	sara_fit@example.com
5	Fahad	AlOtaibi	fahad2025@example.com